Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

AUXILIARY EXAMINATION (2026-27 Odd)

B. E. - Semester IV
September 05, 2026, Saturday, 05:30 PM – 08:30 PM
Time: 3 Hours,
Name Of Faculty: Dr. Saurabh Arora

Course Code: UTA018
Course Name: Object Oriented Programming
M. Marks: 100

Note: *Attempt **any five** questions. Assume missing data, if any, suitably.*

Q1(a)	Explain the following OOP concepts with suitable C++ examples: • Encapsulation • Data abstraction • Objects and classes • Friend function • this pointer	(10)
(b)	Design a C++ class Employee containing employee ID, name and basic salary as data members. Implement: a parameterized constructor, a default constructor, a member function to calculate gross salary, a function to display employee details and constructor overloading for initializing objects in different ways.	(10)
Q2(a)	What is inheritance? Explain single, multiple, multilevel, hierarchical and hybrid inheritance with suitable diagrams/examples. Also explain the significance of public, private and protected inheritance.	(10)
(b)	Develop a C++ program to implement an E-Commerce Product Management System using multilevel inheritance. Consider the following class hierarchy: Product → ElectronicProduct → Laptop The Product class should store the product ID, product name, and price. The ElectronicProduct class should contain the brand name and warranty period. The Laptop class should contain the RAM size and storage capacity. Use appropriate constructors to initialize the data members of all three classes. Implement a function to calculate the final price after applying a given discount and display all product details.	(10)
Q3(a)	What is polymorphism? Explain function overloading, operator overloading and function overriding. Clearly distinguish between compile-time and run-time polymorphism.	(10)
(b)	Write a C++ program to create a class Time containing hours and minutes. Overload the + operator to add two Time objects. The program should correctly convert minutes exceeding 60 into hours.	(10)
Q4(a)	Explain virtual functions and pure virtual functions. Why are virtual functions required for run-time polymorphism? What is an abstract class? Can an object of an abstract class be created directly? Justify your answer.	(10)
(b)	Design an abstract class Employee having a pure virtual function calculateSalary(). Derive two classes: PermanentEmployee, ContractEmployee. For a permanent employee, salary should be calculated using basic salary and allowances. For a contract employee, salary should be calculated using hours worked and hourly rate. Use base-class pointers to demonstrate run-time polymorphism.	(10)

Q5(a)	Explain the C++ exception-handling mechanism using try, throw and catch. Discuss the advantages of exception handling over conventional error handling.	(10)
(b)	Design a class template Calculator<T> that performs addition, subtraction, multiplication and division on two values of type T. Use exception handling to handle division by zero.	(10)
Q6(a)	Explain the following components of the C++ Standard Template Library with suitable examples: • Containers • Iterators • Algorithms Differentiate between vector, list, stack and queue with respect to their organization and common operations.	(10)
(b)	Write a C++ program to maintain the marks of students using an STL vector. The program should: 1. Accept marks of n students. 2. Display all marks using an iterator. 3. Find the highest and lowest marks using STL algorithms. 4. Sort the marks in ascending order. 5. Search for a specified mark.	(10)

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